

Connection Diagrams



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Control Accessories - General

Pressure Reducing (PR):

Regulating the valve downstream (outlet) pressure to a fixed and user predetermined set-point value.

Valve closure will result a pressure decrease.

Pressure Sustaining (PS):

Regulating the valve upstream (inlet) pressure to a fixed and user predetermined set-point value.

Valve closure will result a pressure increase.

3-way Pilot Valves (Pages 12-14):

Ports Marking:

S - Pressure sensing port

P - Port for connection of upstream pressure

C - Command port, connected to the valve control chamber

V - Vent port. Through this port water are vented when the valve opens

Operating Principle:

CLOSE - Connects ports P to C thus letting the upstream pressure into the valve bonnet

OPEN - Connects ports C to V thus venting the pressure in the valve bonnet

LOCK - Keep the valve in the present position by blocking all ports

2-way Pilot Valves (Pages 15-18):

Operating Principle:

Increases or decreases a drainage passage from the valve control chamber to downstream. The inlet to the chamber is via a fixed orifice or a regulated needle valve.

CLOSE - The drainage flow through the pilot is smaller than the inlet flow

OPEN - The drainage flow through the pilot is larger than the inlet flow

LOCK - The drainage flow through the pilot is equal to the inlet flow

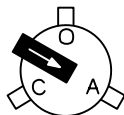
3-way Manual Selector Valve:

A ball valve with one common (fixed) port and three functional ports:

"CLOSE" - Connected to the upstream

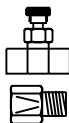
"OPEN" - Vented to the atmosphere

"AUTO" - Connected to the command port of the control device (pilot, solenoid etc.)



Needle Valve / Orifice:

For flow limitation in the control system and reduction of the response speed of the main valve.



Shuttle Valve:

Tee shaped pressure selector - selecting the higher pressure when the valve is controlled by two pressure sources.



3-way Solenoid:

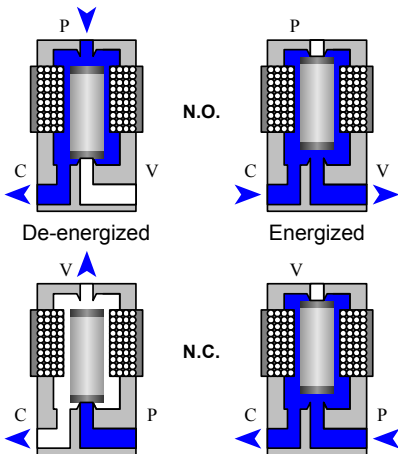
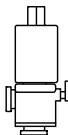
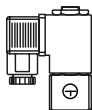
Converts electric pulses to an hydraulic command.

Ports Marking:

P - Pressure port. Connected to the upstream pressure

C - Connected to the control chamber of the valve

V - Pressure venting port



Relays

- Allow fast response of remote controlled valves.
- Acceleration of hydraulic commands.
- Converting N.O-N.C characteristics.
- Enable overcoming topographic elevation differences.

Ports Marking:

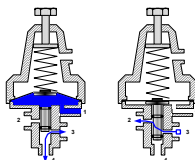
S - Hydraulic command inlet port

P - Upstream pressure connection port

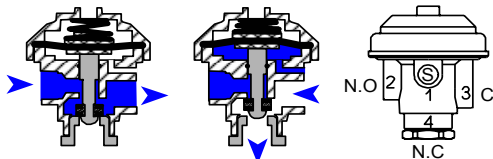
C - Connected to the control chamber of the valve

V - Pressure venting port

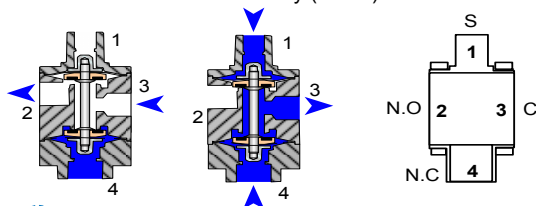
25-100 - 3-way (Plastic)



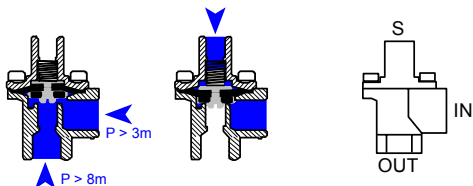
66-210 - 3-way (Brass)



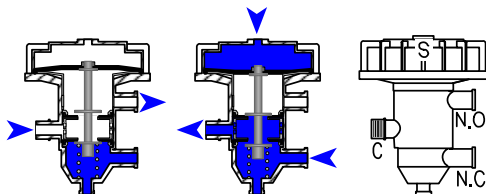
28-300 - 3-way (Brass)



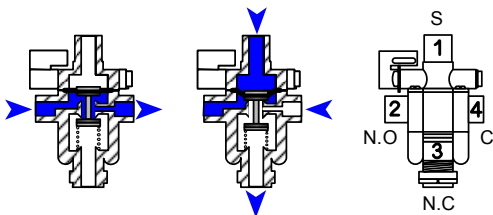
28-200 - 2-way (Brass)



25-300 - 3-way (Plastic)



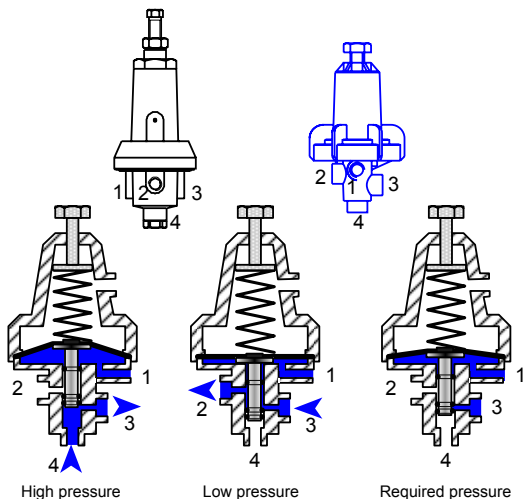
Galit - 3-way (Plastic)



3-way Pilots

31-310 (Brass)

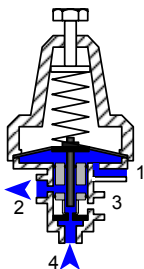
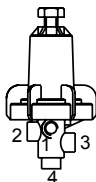
29-200 (Plastic)



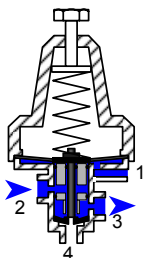
CONNECTIONS

PORT	PRESSURE REDUCING	PRESSURE SUSTAINING
1	Downstream	Upstream
2	Vent	Upstream
3	Valve Chamber	Valve Chamber
4	Upstream	Vent

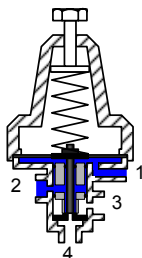
29-100 (Plastic)



High pressure



Low pressure



Required pressure

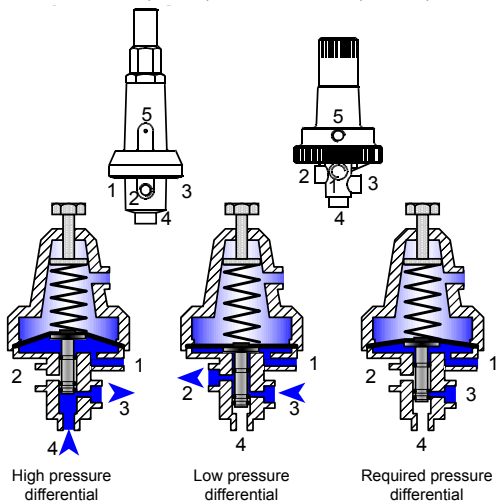
CONNECTIONS

PORT	PRESSURE REDUCING
1	Downstream
2	Valve Chamber
3	Vent
4	Upstream

3-way Pilots

76-200 (Brass)

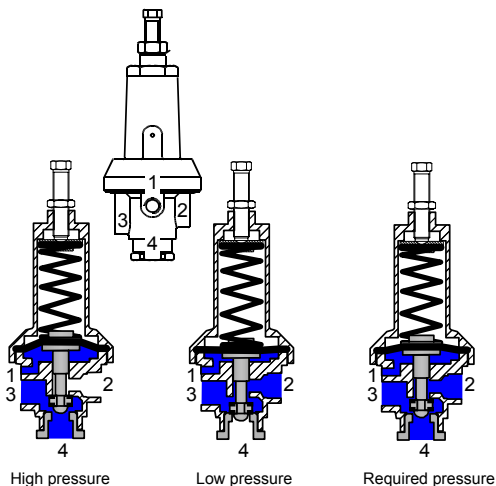
29-300 (Plastic)



CONNECTIONS

PORT	P. D. REDUCING	P. D. SUSTAINING
1	High Pressure	High Pressure
2	Vent	Upstream
3	Valve Chamber	Valve Chamber
4	Upstream	Vent
5	Low Pressure	Low Pressure

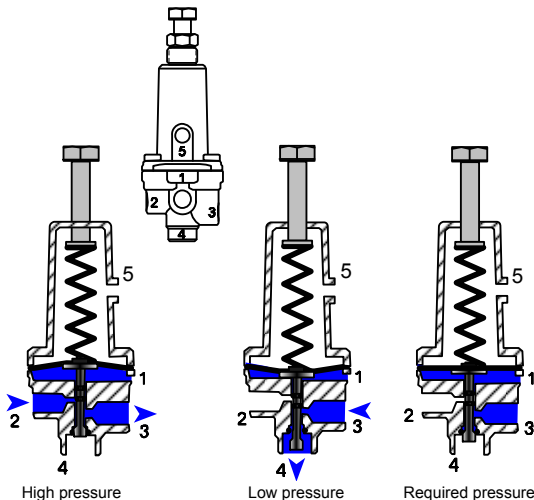
66-310 / 68-710 (Brass)



CONNECTIONS

PORT	66-310 / 68-710 P. RELIEF	66-310: RELAY
1	Upstream	Command
2	Plugged	N.O. Port
3	Valve Chamber	Valve Chamber
4	Vent	N.C. Port

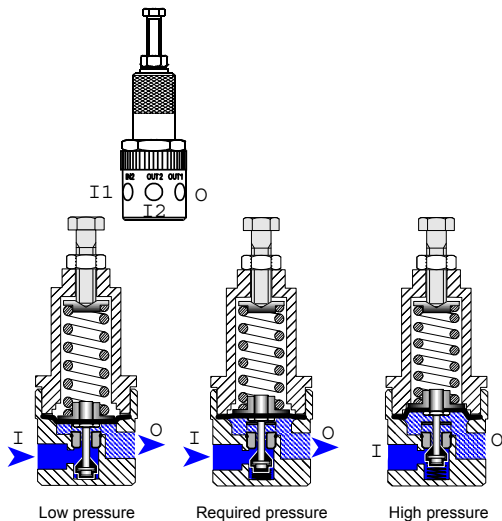
31-100 / 31-10D (Brass)



CONNECTIONS

PORT	P. REDUCING	P.D. REDUCING
1	Downstream	High Pressure
2	Upstream	Upstream
3	V. Chamber	V. Chamber
4	Vent	Vent
5	-	Low Pressure

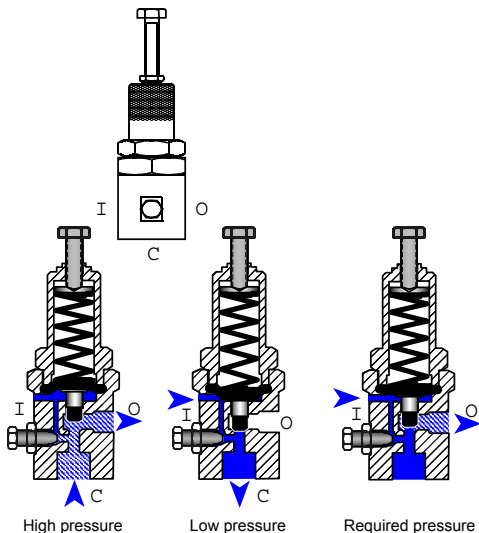
68-410 (Brass)



CONNECTIONS

PORT	PRESSURE REDUCING
I1	Valve Control Chamber
I2	Upstream (Through Orifice)
O	Downstream

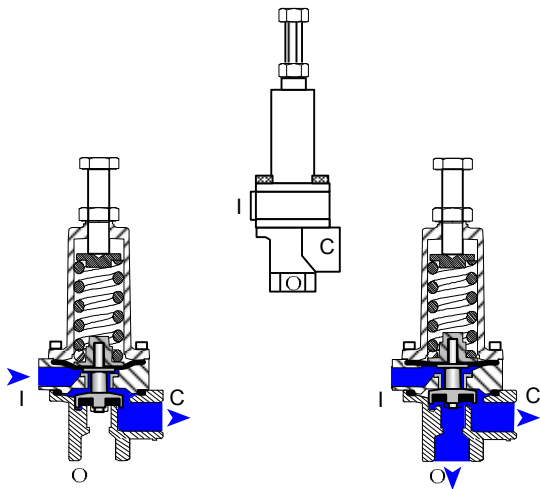
68-500 (Brass)



CONNECTIONS

PORT	PRESSURE SUSTAINING
I	Upstream
C	Valve Control Chamber
O	Downstream

68-220 (Brass)

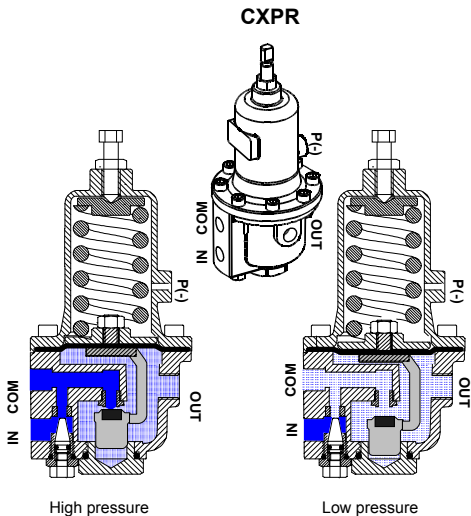


Low pressure

High pressure

CONNECTIONS

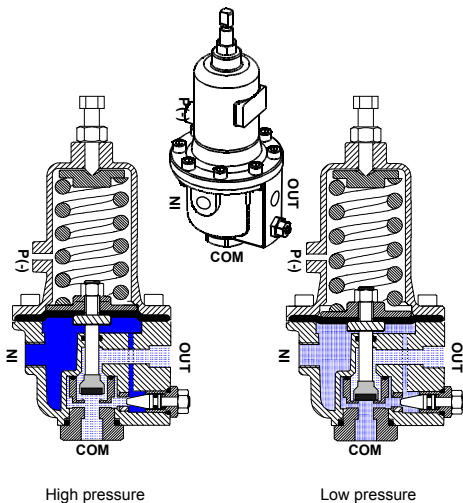
PORT	PRESSURE RELIEF
I	Upstream
C	V. Chamber
O	Downstream / Vent



CONNECTIONS

PORT	CXPR PRESSURE REDUCING
IN	Upstream
OUT	Downstream
COM	Valve Chamber
P (-)	Not Connected

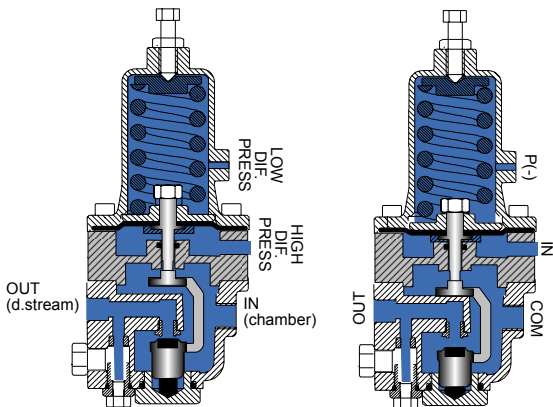
CXPS / CXSD (BRASS)



CONNECTIONS

PORT	CXPS P. SUSTAINING	CXSD DIFF. P. SUSTAINING
IN	Upstream	Upstream
OUT	Downstream	Downstream
COM	Valve Chamber	Valve Chamber
P (-)	Not Connected	Low Ref. Pressure

CXRS



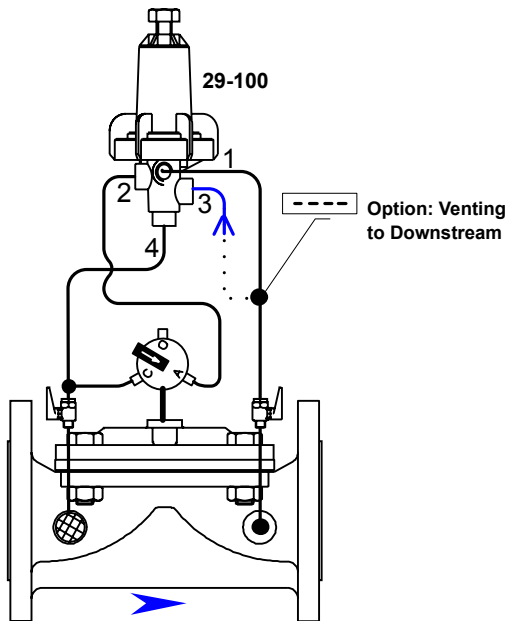
Excessive Flow-Rate

Deficient Flow-Rate

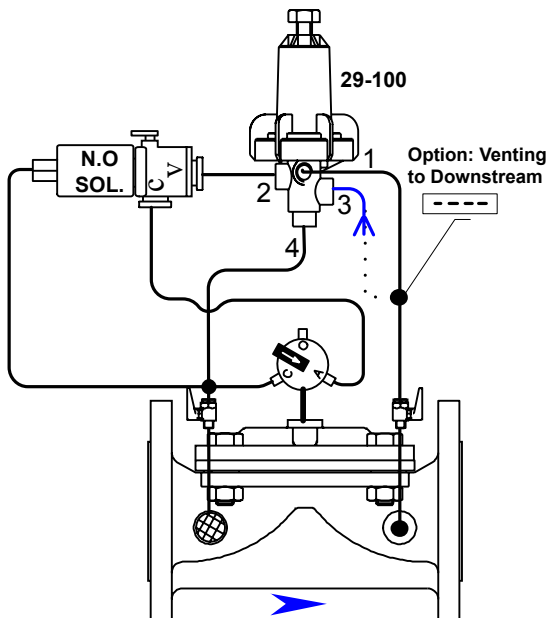
CONNECTIONS

PORT	CXRS PRESSURE REDUCING
IN	Upstream Orifice
OUT	Downstream
COM	Control Chamber
P (-)	Downstream Orifice

Pressure Reducing Valve (PR) Using 29-100 Mini Pilot (Black)

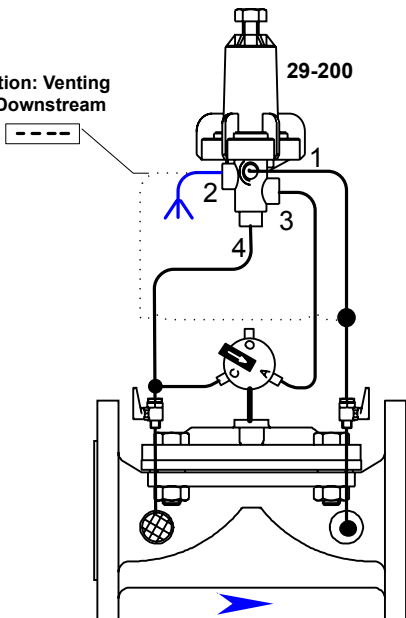


Pressure Reducing Electric Valve (PR/EL) Using 29-100 Mini Pilot (Black)



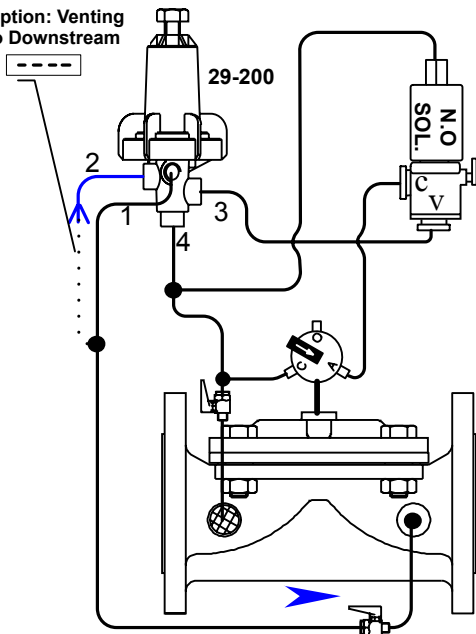
Pressure Reducing Valve (PR) Using 29-200 Mini Pilot (Blue)

Option: Venting
to Downstream

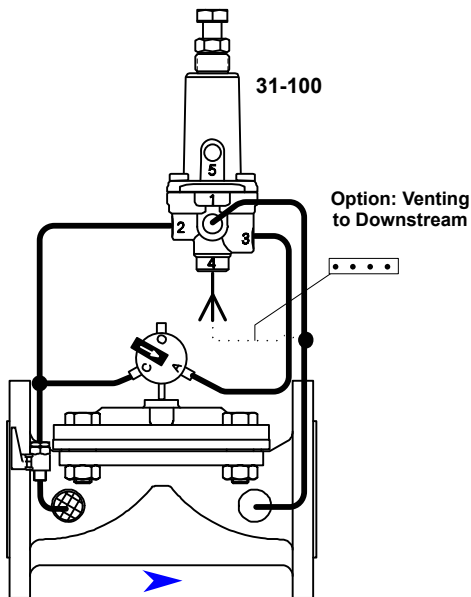


Pressure Reducing Electric Valve (PR/EL) Using 29-200 Mini Pilot (Blue)

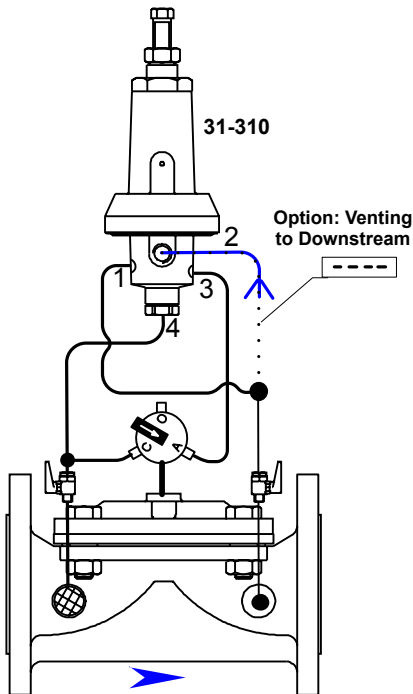
Option: Venting
to Downstream



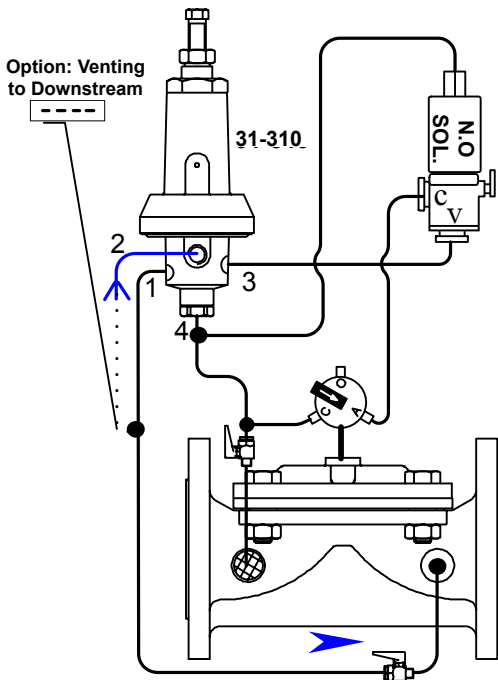
Pressure Reducing Valve (PR) Using 31-100 Mini Pilot (3w-brass)



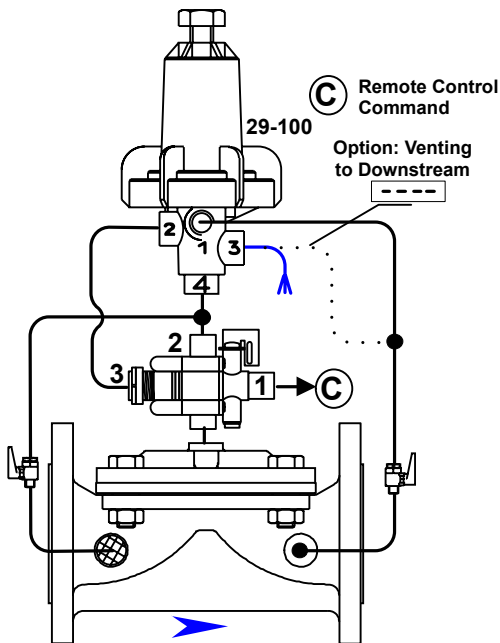
Pressure Reducing Valve (PR) Using 31-310 Pilot (Brass)



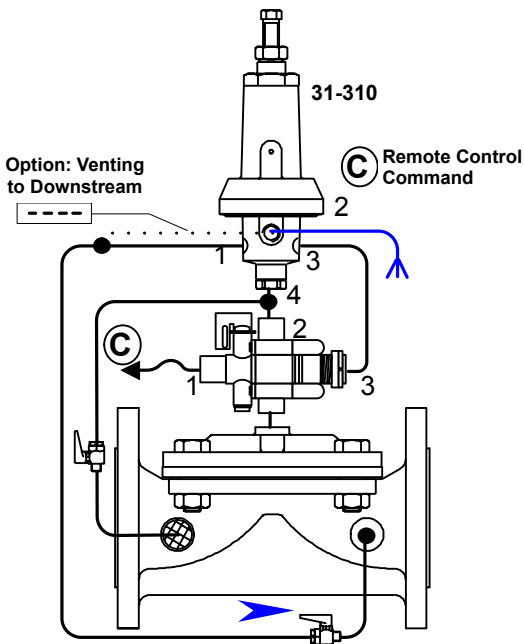
Pressure Reducing Electric Valve (PR/EL) Using 31-310 Pilot (Brass)



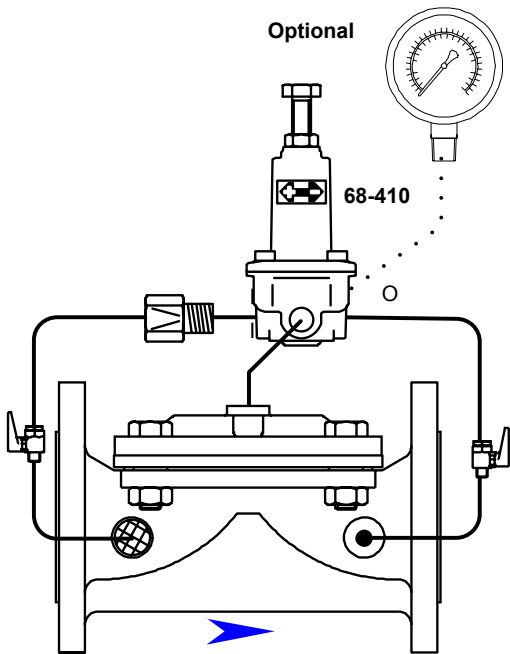
Pressure Reducing / Remote Control Valve (PR/RC) Using 29-100 Mini Pilot (Black) and “Galit”



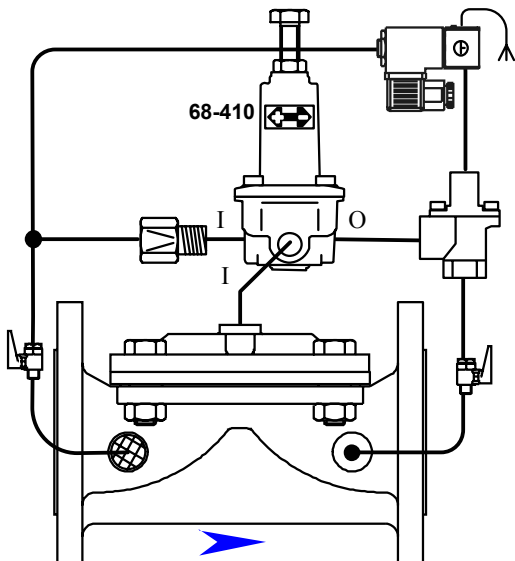
Pressure Reducing / Remote Control Valve (PR/RC) Using 31-310 Pilot (Brass) and “Galit”



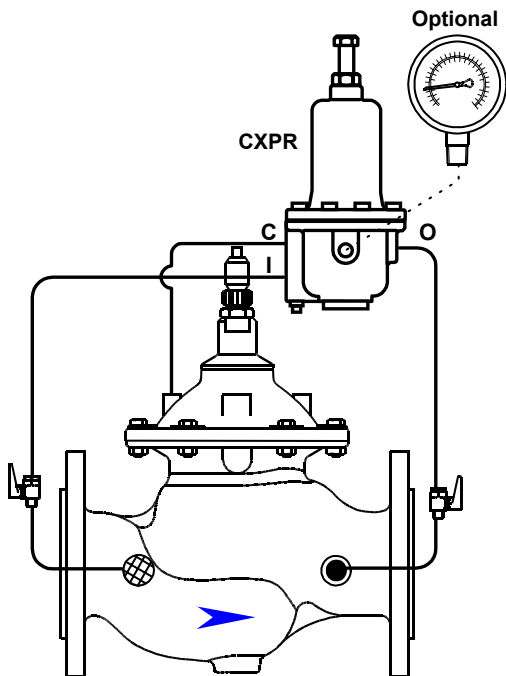
Pressure Reducing Valve (PR) Using 68-410 Pilot (2w-brass)



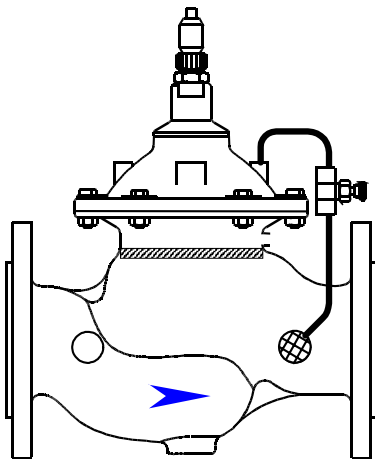
Pressure Reducing Electric Valve (PR/EL) Using 68-410 Pilot (2w-brass)



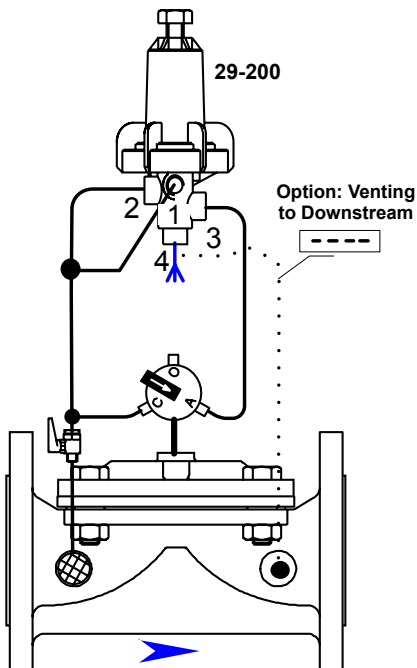
Pressure Reducing Valve (PR) Using CXPR Pilot (2w-brass)



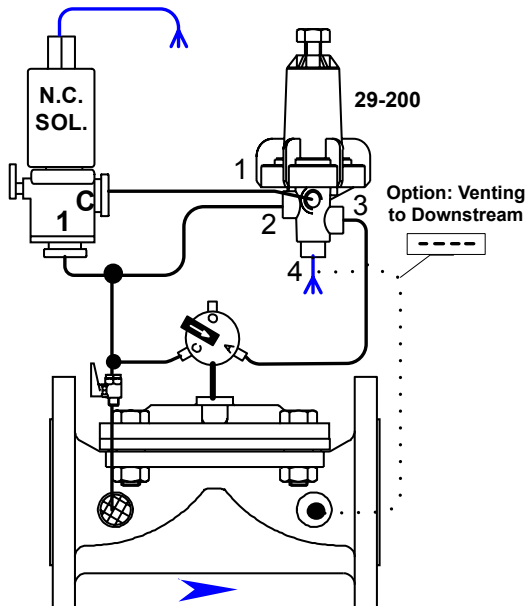
Proportional Pressure Reducing Valve (PRD)



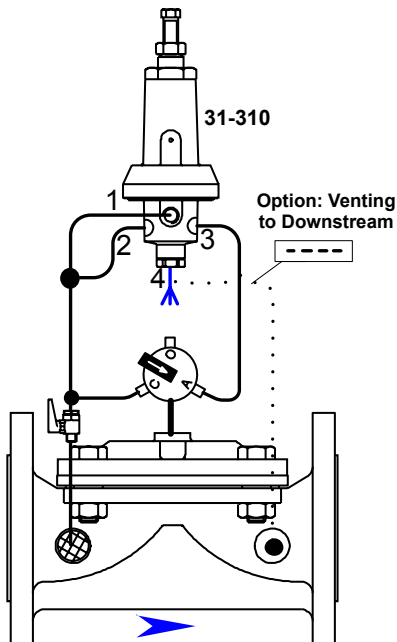
Pressure Sustaining Valve (PS) Using 29-200 Mini Pilot (Blue)



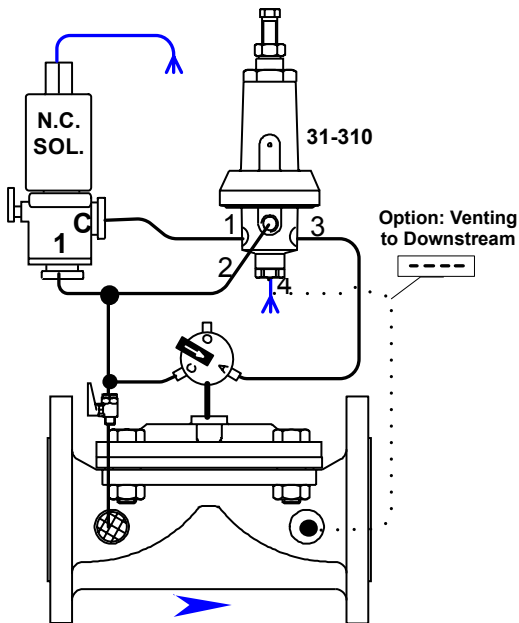
Pressure Sustaining/Electric Valve (PS/EL) Using 29-200 Mini Pilot (Blue)



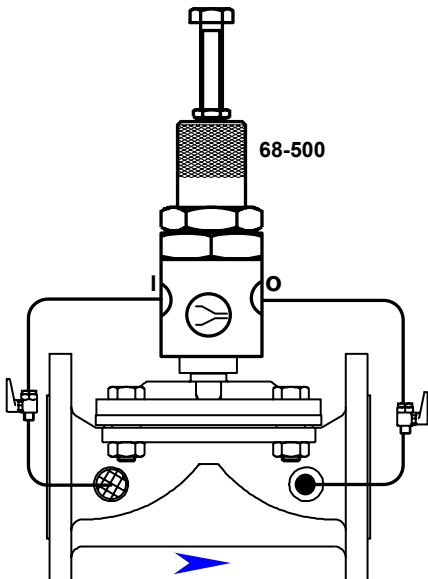
Pressure Sustaining Valve (PS) Using 31-310 Pilot (Brass)



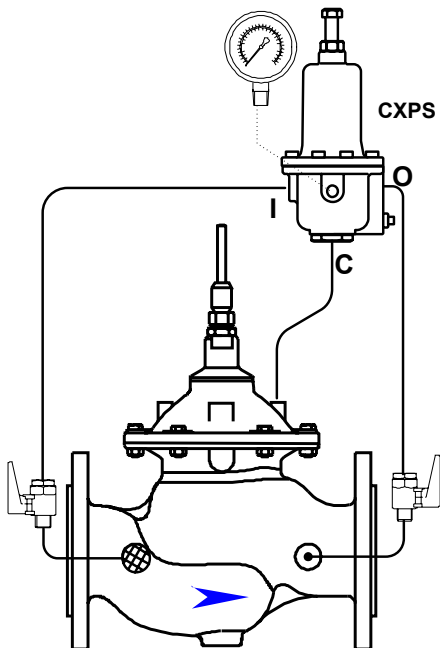
Pressure Sustaining/Electric Valve (PS/EL) Using 31-310 Pilot



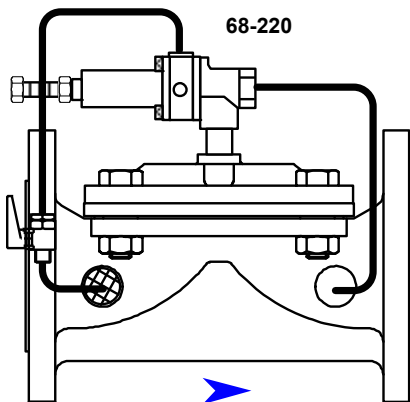
Pressure Sustaining Valve (PS) Using 68-500 Pilot (2-way)



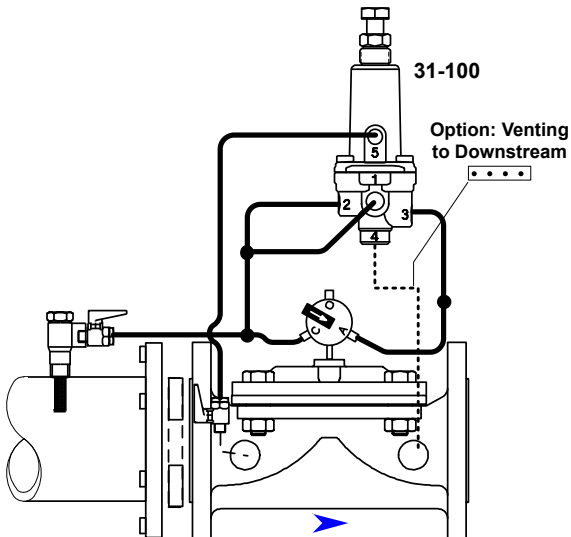
Pressure Sustaining Valve (PS) Using CXPS Pilot (2-way)



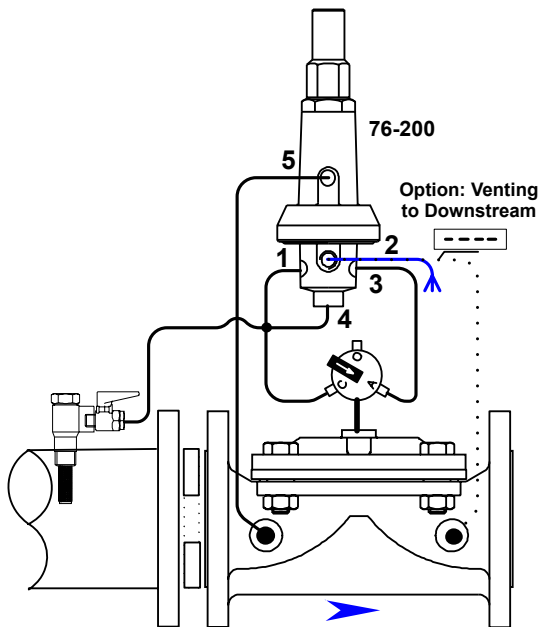
Quick Relief Valve (QR) Using 68-220 Pilot (2-way)



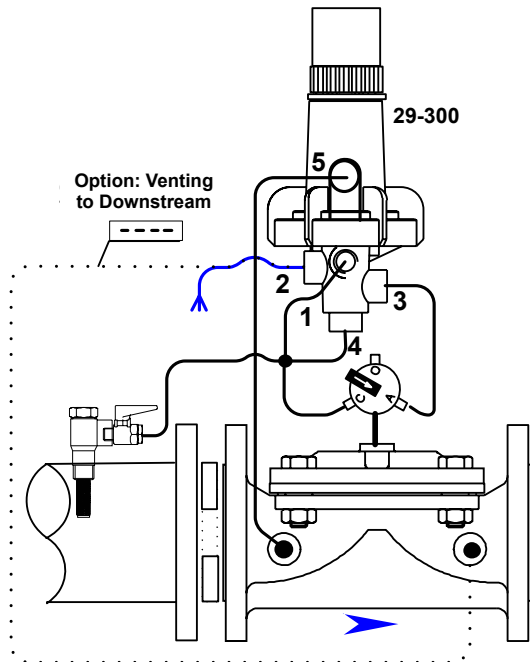
Flow Rate Control Valve (FR)
Using 31-10D Mini Pilot (3w-brass)



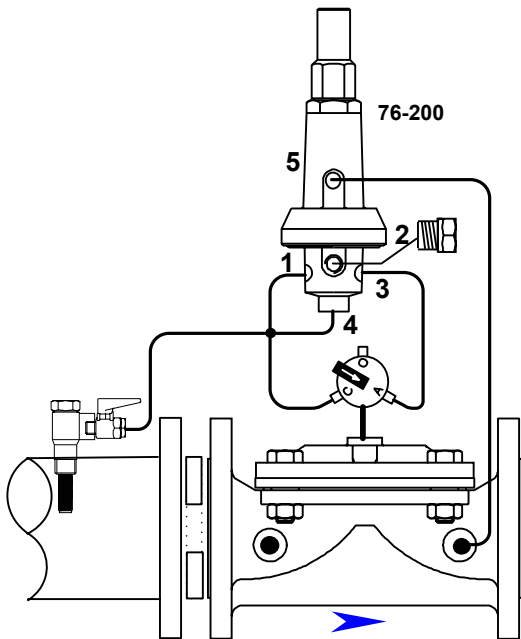
Flow Rate Control Valve (FR) Using 76-200 Differential Pilot



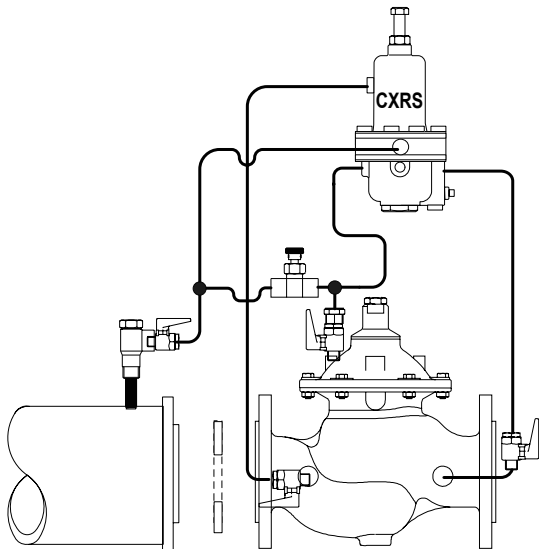
Flow Rate Control Valve (FR) Using 29-300 Differential Pilot



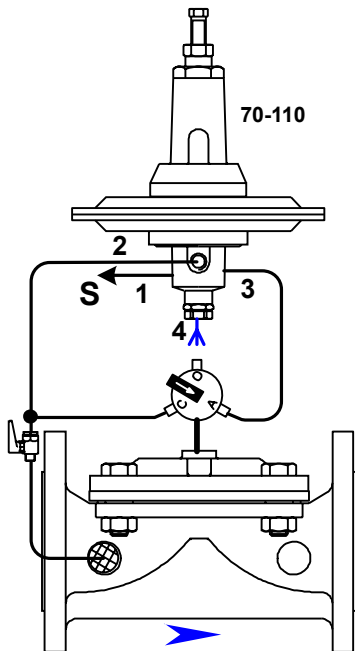
Excessive Flow Shut-off Valve (FE) Using 76-200 Differential Pilot



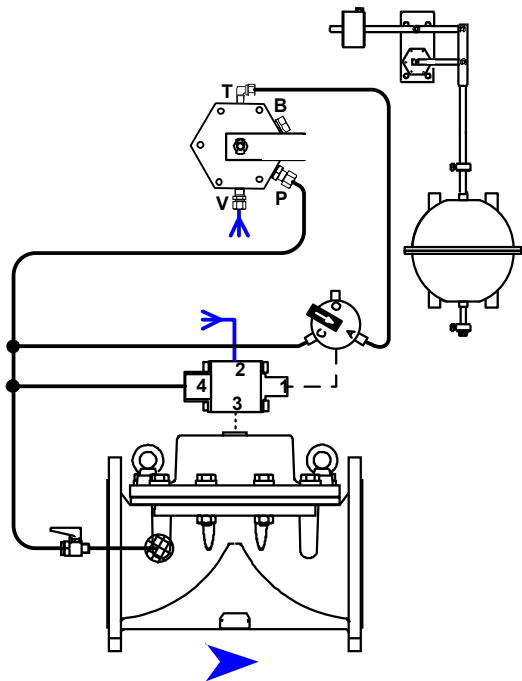
Flow Rate Control Valve (FR) Using CXRS Differential Pilot



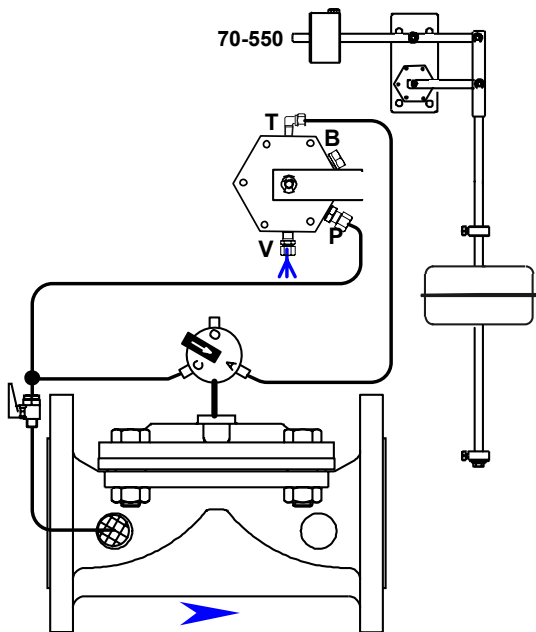
High Level Control Valve (AL) Using 70-110 Pilot



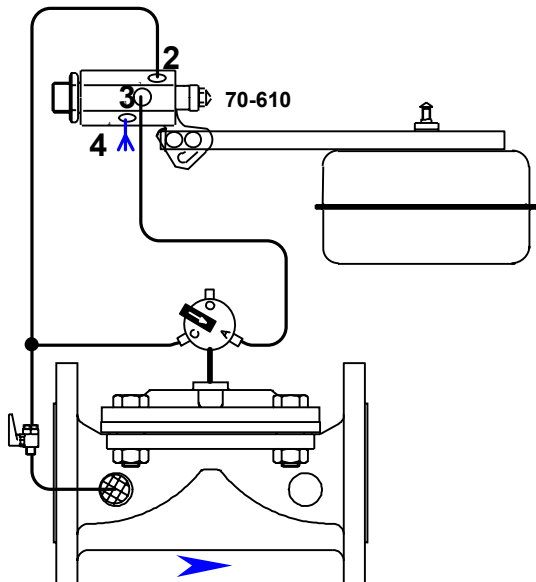
Level Control (FLDI) for Large Valves Using 70-550 Differential Float Pilot and 28-300 Relay



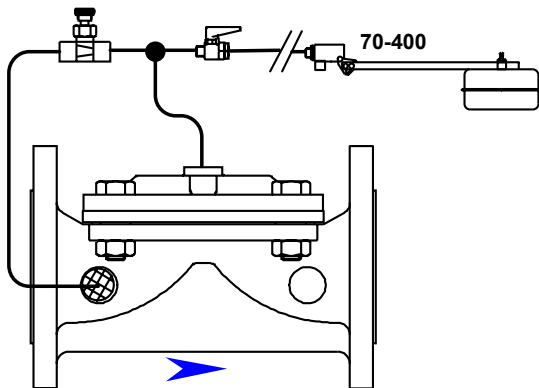
Level Control Valve (FLDI) Using 70-550 Differential Float Pilot



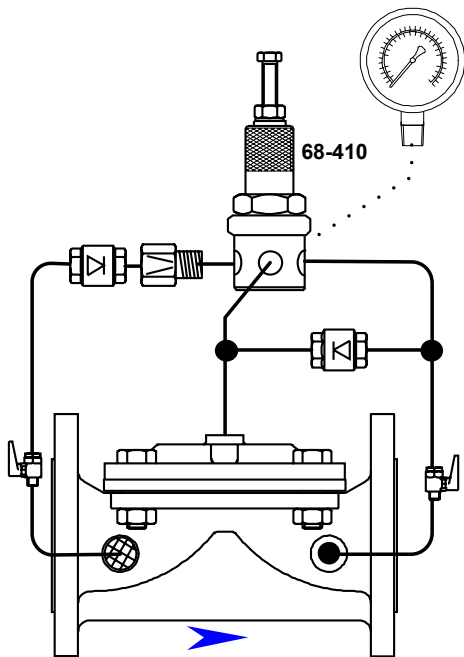
Level Control Valve (FLDI) Using 70-610 Differential Float Pilot



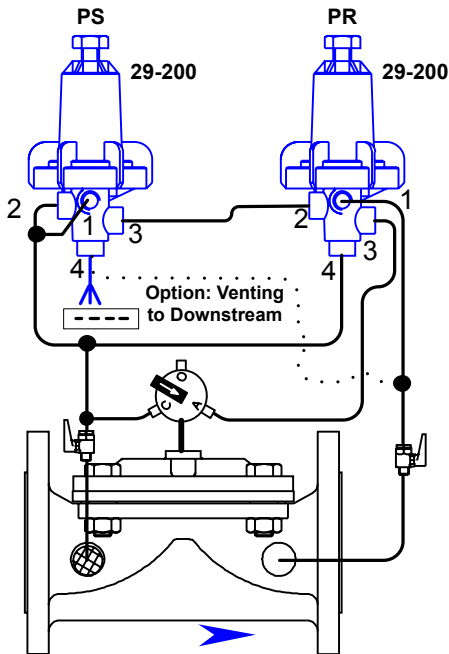
Level Control Valve (FL) Using 70-400 Float Pilot



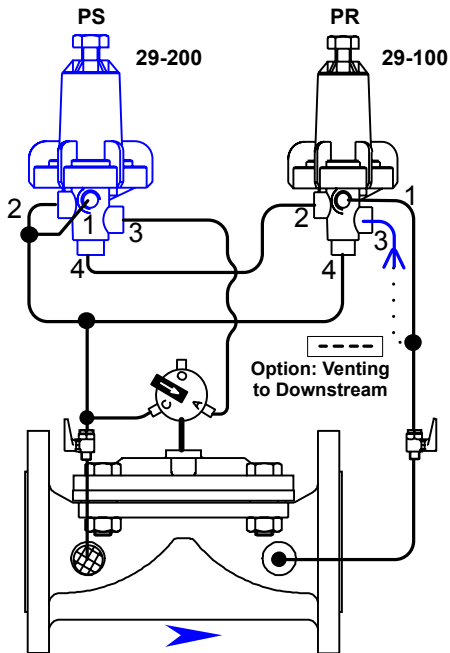
Pressure Reducing and Check Valve (PR/CV) Using 68-410 Pilot (2 Way)



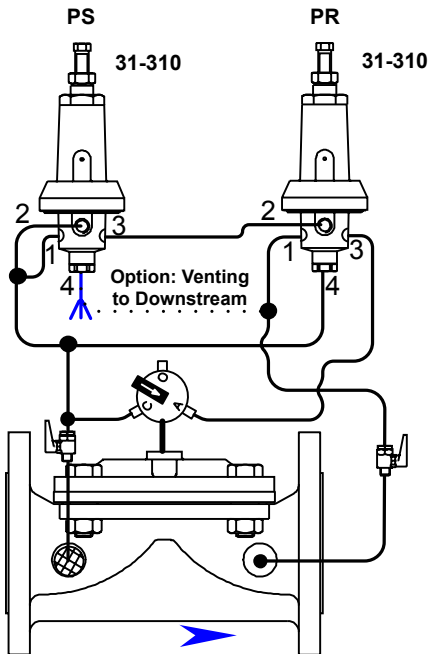
Pressure Reducing and Sustaining Valve (PR/PS) Using two 29-200 Mini Pilots (Blue)



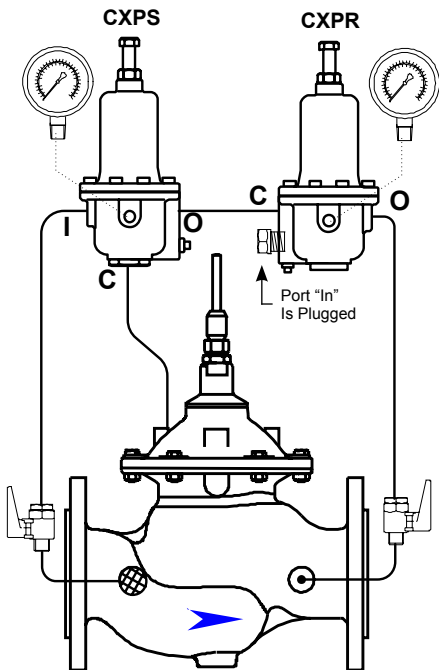
Pressure Reducing and Sustaining Valve (PR/PS) Using 29-100 Mini Pilot (Black) + 29-200 Mini Pilot (Blue)



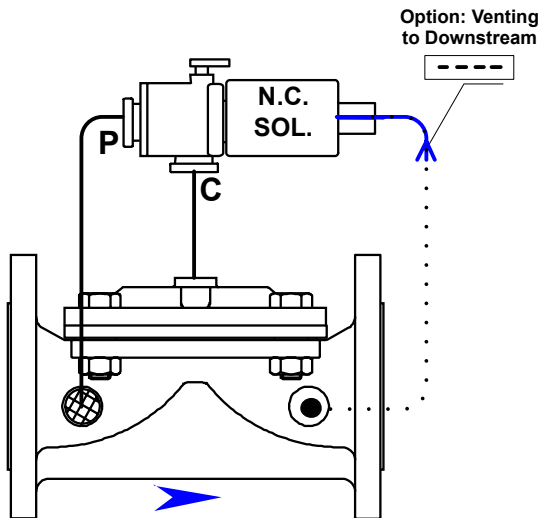
Pressure Reducing and Sustaining Valve (PR/PS) Using two 31-310 Pilots (Brass)



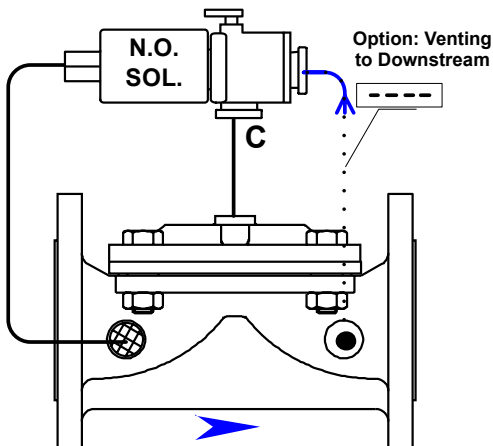
Pressure Reducing and Sustaining Valve (PR/PS) Using CXPS and CXPR Pilots



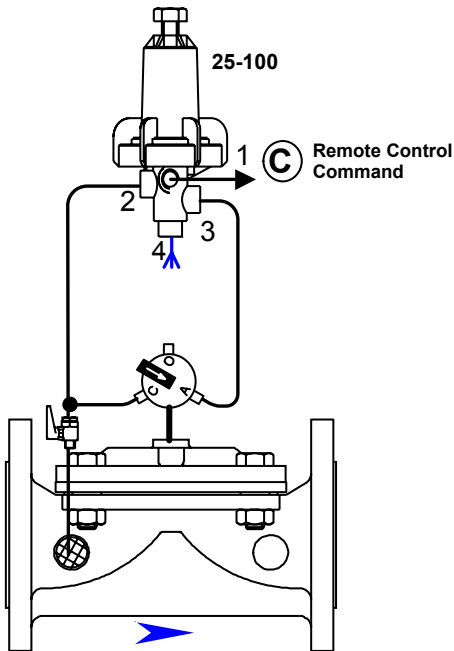
N.O. Electric Valve (EL)



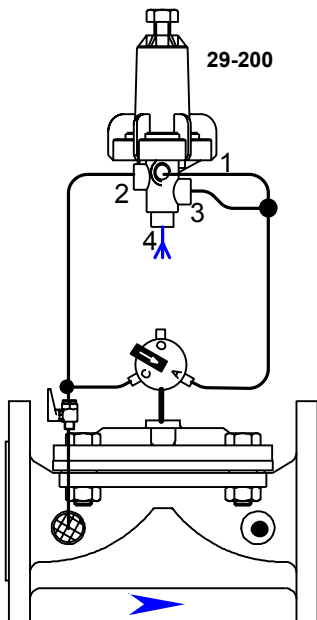
N.C. Electric Valve (EL)



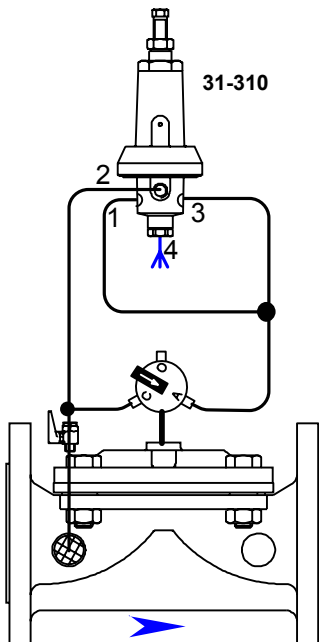
Topography Diff. Compensation Remote Control Valve (RC) Using 25-100 Mini Pilot (Blue)



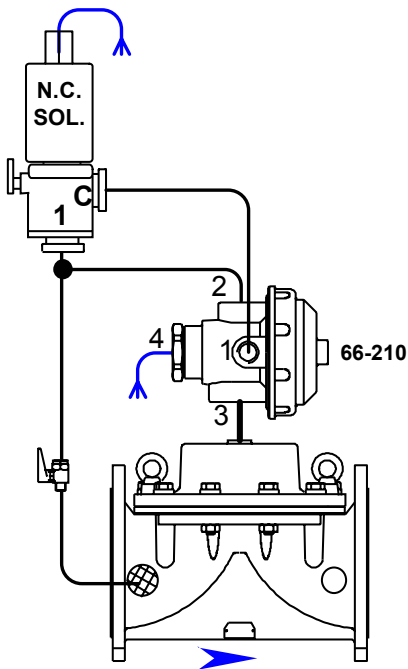
Two Stage Opening (TO) Using 29-200 Mini Pilot (Blue)



Two Stage Opening (TO) Using 31-310 Pilot (Brass)

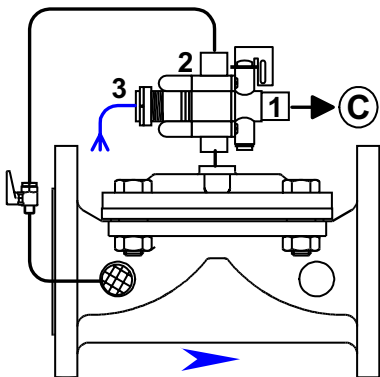


N.C. Electric Valve (EL) Using 66-210 Accelerator Relay

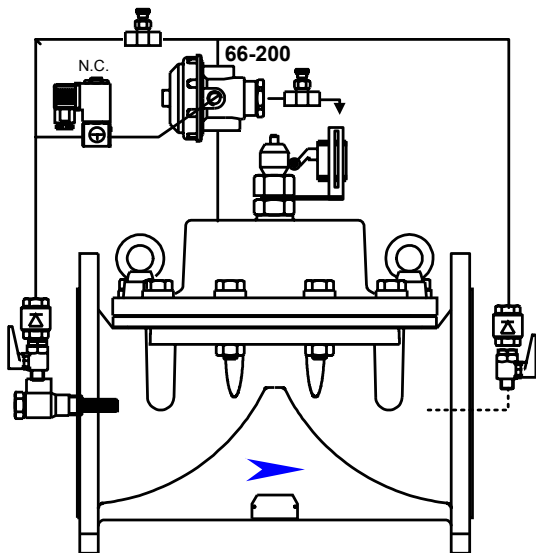


Remote Control Valve (RC) Using Galit

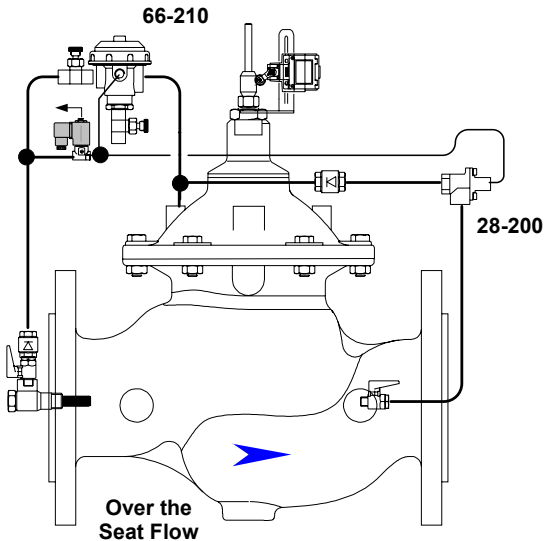
(C) Remote Control Command



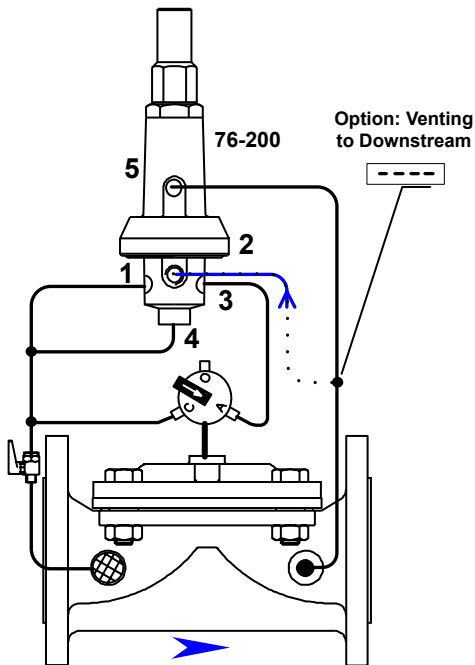
Pump Control Valve (BC) with S-100



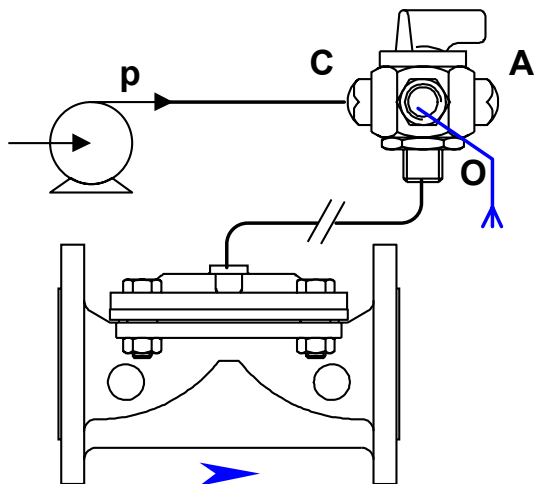
Pump Control Valve (BC) with S-300



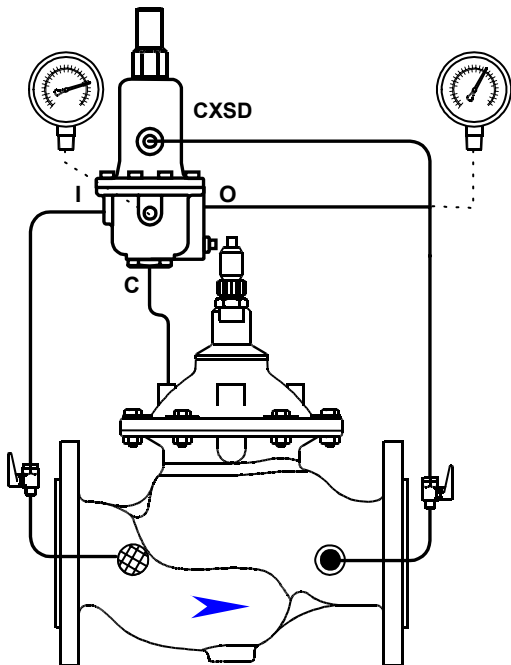
Differential Pressure Control Valve (DI) Using 76-200 Differential Pilot



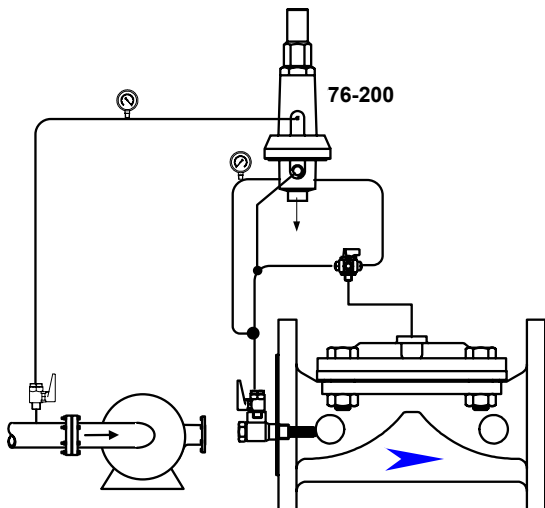
Remote Control Valve (RC) By Command Tube



Differential Pressure Sustaining Valve (DI) Using CXSD Differential Pilot



Differential Pressure Sustaining Valve (DI) Using 76-200 Differential Pilot





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